



**MACHINE LEARNING BASED  
BUYER SEGMENTATION AND  
INVESTMENT PROFILING FOR  
REAL ESTATE MARKET  
INTELLIGENCE**

**ISRAR SIDDIQUI**

# Machine learning based buyer segmentation and investment profiling for real estate market intelligence

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# MACHINE LEARNING BASED BUYER SEGMENTATION AND INVESTMENT PROFILING FOR REAL ESTATE MARKET INTELLIGENCE

## Internship Project Report

Submitted as part of the Business Analyst Internship at Unified Mentor

Internship Organization: Unified Mentor

Internship Domain: Business Analyst

Project Domain: Financial Analytics & Real Estate Market Intelligence

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## TITLE PAGE

# MACHINE LEARNING BASED BUYER SEGMENTATION AND INVESTMENT PROFILING FOR REAL ESTATE MARKET INTELLIGENCE

A Project Report Submitted During

BUSINESS ANALYST INTERNSHIP

At

UNIFIED MENTOR

Internship Domain: Business Analyst

This project focuses on applying business analysis, data analytics and machine learning techniques to understand real estate buyer behaviour, identify customer segments and develop investment-oriented buyer profiles.

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## CERTIFICATE

This is to certify that the project entitled “Machine Learning Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence” was undertaken as part of the Business Analyst Internship at Unified Mentor.

The project focuses on analysing real estate customer data, identifying meaningful customer segments using machine learning, and converting the analytical results into business insights that can support customer targeting, marketing and investment profiling.

The work demonstrates the practical application of business analysis, data interpretation, machine learning, visualization and business intelligence concepts.

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## DECLARATION

I hereby declare that this project report entitled “Machine Learning Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence” has been prepared as part of my Business Analyst internship at Unified Mentor.

The report is based on the project datasets, analytical outputs and project requirements provided for the study. The analysis has been performed using data preparation, exploratory analysis, machine learning-based clustering and visualization techniques.

The findings presented in this report are intended for academic and project purposes.

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## ACKNOWLEDGEMENT

I would like to sincerely thank Unified Mentor for providing me with the opportunity to work on this project as part of my Business Analyst internship.

This internship gave me an opportunity to apply the concepts of business analysis and data analytics to a practical business problem. Working on the real estate buyer segmentation project helped me understand how raw business data can be transformed into useful information for decision-making.

I am grateful to the mentors and everyone who provided guidance and support during the project. I also appreciate the resources and datasets made available for completing the analysis.

The project helped me improve my understanding of data cleaning, exploratory data analysis, customer segmentation, machine learning, dashboard development and business interpretation.

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## EXECUTIVE SUMMARY

The real estate industry deals with customers having different needs, purchasing purposes, financial requirements and behavioural characteristics. Because of this diversity, treating every customer in the same way may not always be effective.

The purpose of this project is to use data analytics and machine learning to understand real estate buyers and divide them into meaningful customer groups.

The project, titled “Machine Learning Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence,” was completed during a Business Analyst internship at Unified Mentor.

The analysis uses customer information related to demographics, geography, acquisition purpose, financing behaviour, referral channels and satisfaction. The project requirements identify these variables as important factors for understanding real estate buyers.

The analytical application successfully processed 2,000 customer records and 20 features, with 3 clusters selected for customer segmentation.

K-Means clustering was used as the main machine learning technique. The resulting model achieved a Silhouette Score of 0.4382, which was interpreted by the application as moderate clustering quality.

The resulting customer groups provide useful profiles for business analysis. The analysis can help a real estate organization understand different types of buyers, identify investment-oriented customers, understand financing behaviour and develop more focused marketing strategies.

The project also includes dashboard-based visualization through Streamlit and Tableau, allowing analytical results to be presented in a more understandable form.

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## CHAPTER 1: INTRODUCTION

### 1.1 Introduction

Business decisions are increasingly influenced by data. Organizations collect large amounts of information about their customers, transactions, products and markets, but the real value of this information comes from being able to interpret it and use it effectively.

The real estate industry is a good example of a business environment where customer data can provide valuable insights. Different buyers may purchase properties for different reasons. Some may be looking for a home, while others may be interested in investment. Customers may also differ in age, location, financing requirements and preferred channels of communication.

For a real estate organization, understanding these differences is important for effective customer targeting.

This project applies business analysis and machine learning techniques to identify groups of customers with similar characteristics. The resulting customer segments can then be interpreted from a business perspective.

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## 1.2 Project Overview

The project focuses on buyer segmentation and investment profiling in the real estate market.

The main analytical process involves:

Data → Cleaning → Encoding → Scaling → Exploratory Analysis → Clustering → Evaluation → Buyer Profiling → Business Insights

The primary machine learning method used in the project is K-Means clustering.

The Streamlit application describes the project as a machine-learning-based customer segmentation system using K-Means clustering.

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## CHAPTER 2: INTERNSHIP OVERVIEW

### 2.1 About the Internship

This project was completed as part of a Business Analyst internship at Unified Mentor.

The internship provided an opportunity to work with business-oriented data and understand how analytical techniques can support decision-making.

As a Business Analyst, one important responsibility is to understand a business problem, analyse relevant data and communicate findings in a way that can be understood by decision-makers.

This project follows the same approach.

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### 2.2 Internship Domain

Business Analyst

The project involves several activities relevant to business analysis, including:

- \* Understanding business requirements
  - \* Understanding the available data
  - \* Data cleaning
  - \* Data analysis
  - \* Identifying patterns
  - \* Customer segmentation
  - \* Interpreting analytical results
  - \* Developing business insights
  - \* Preparing recommendations
  - \* Creating dashboards and visualizations
- 

## 2.3 Learning from the Internship

The project provided practical exposure to:

- \* Business problem analysis
  - \* Data-driven decision-making
  - \* Customer behaviour analysis
  - \* Machine learning concepts
  - \* Data visualization
  - \* Dashboard interpretation
  - \* Business reporting
  - \* Communication of analytical findings
- 

## CHAPTER 3: PROJECT BACKGROUND

### 3.1 Background

Real estate businesses deal with a wide range of customers. A customer looking for a residential property may have completely different requirements from an investor looking for properties as part of an investment strategy.

Similarly, customers may differ in their need for financing, preferred acquisition channels and geographical preferences.

The project requirement document highlights the need for a data-driven understanding of:

- \* Different types of property buyers
- \* Investment motivations
- \* Geographic differences in investment behaviour
- \* Customer financing patterns

Without this understanding, businesses may spend marketing resources on broad campaigns that do not address the specific needs of individual customer groups.

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### 3.2 Role of Customer Segmentation

Customer segmentation divides a large customer base into smaller groups based on similarities.

For example, customers may be grouped according to:

- \* Age
- \* Customer type
- \* Location
- \* Investment purpose
- \* Financing behaviour
- \* Acquisition channel
- \* Satisfaction

Machine learning can make this process more systematic by identifying patterns across multiple variables simultaneously.

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## CHAPTER 4: PROBLEM STATEMENT

The main problem addressed by this project is the difficulty of understanding diverse real estate customers using traditional, generalized approaches.

A business may have thousands of customers but may not know:

- \* Which customers are investment-oriented
- \* Which customers are more likely to require financing
- \* Which customer groups have similar characteristics
- \* Which customer groups should receive different marketing messages
- \* How customer characteristics differ across locations

Therefore, the central problem can be stated as:

How can machine learning and business analytics be used to identify meaningful real estate buyer segments and develop investment-oriented customer profiles?

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## CHAPTER 5: OBJECTIVES

The major objectives of the project are:

### 5.1 Customer Segmentation

To divide real estate customers into meaningful groups using machine learning.

### 5.2 Buyer Behaviour Analysis

To understand differences between customer groups based on demographic and behavioural characteristics.

### 5.3 Investment Profiling

To identify customer groups with stronger investment-oriented characteristics.

### 5.4 Financing Analysis

To understand loan application behaviour across customer segments.

### 5.5 Geographic Analysis

To analyse customer distribution across countries and regions.

### 5.6 Business Intelligence

To convert analytical findings into practical business insights.

### 5.7 Visualization

To present the results using dashboards and charts that can support business decision-making.

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## CHAPTER 6: SCOPE OF THE PROJECT

The scope of this project includes:

- \* Customer data analysis
- \* Data cleaning
- \* Feature encoding
- \* Feature scaling
- \* Exploratory data analysis
- \* Customer segmentation
- \* K-Means clustering
- \* Cluster evaluation
- \* Investment profiling
- \* Financing behaviour analysis
- \* Geographic analysis
- \* Property data analysis
- \* Dashboard development
- \* Business recommendations

The project does not attempt to establish causal relationships between customer behaviour and property outcomes.

It also does not focus on predicting property prices, loan defaults or future customer behaviour.

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CHAPTER 7: DATA AND PROJECT REQUIREMENTS

7.1 Customer Dataset

The project dataset contains customer-level information.

The Streamlit analytical application reports:

| Parameter         | Value |
|-------------------|-------|
| Customer Records  | 2,000 |
| Features          | 20    |
| Selected Clusters | 3     |

—

7.2 Major Variables

Important variables include:

| Variable            | Business Meaning                 |
|---------------------|----------------------------------|
| Client ID           | Unique customer identifier       |
| Client Type         | Individual or corporate customer |
| Gender              | Customer gender                  |
| Country             | Customer country                 |
| Region              | Customer geographical region     |
| Age                 | Customer age information         |
| Acquisition Purpose | Home/personal use or investment  |
| Loan Applied        | Whether financing was requested  |
| Referral Channel    | Customer acquisition source      |
| Satisfaction Score  | Customer satisfaction rating     |

These variables provide a combination of demographic, geographic, behavioural and financial information.

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CHAPTER 8: DATA PREPARATION

Data preparation is an important part of the project because the quality of the input data directly affects the quality of the analysis.

8.1 Data Cleaning

The data preparation process includes:

- \* Checking for missing values
- \* Removing duplicate records where required
- \* Standardizing categorical values
- \* Checking data types
- \* Preparing variables for analysis

The goal is to make the dataset consistent and suitable for further analysis.

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## 8.2 Feature Encoding

Several variables in the dataset are categorical.

Examples include:

- \* Client type
- \* Country
- \* Region
- \* Acquisition purpose
- \* Referral channel

These variables need to be converted into numerical representations before they can be used effectively in clustering.

Encoding techniques such as one-hot encoding and label encoding can be used depending on the nature of the variable.

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## 8.3 Feature Scaling

K-Means uses distance calculations to form clusters.

If one variable has a much larger numerical range than another, it can have a disproportionate effect on the clustering process.

Therefore, numerical features are scaled before clustering.

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## CHAPTER 9: EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis was performed to understand the customer dataset before applying the machine learning model.

The main areas examined were:

- \* Customer type
  - \* Gender
  - \* Age
  - \* Acquisition purpose
  - \* Loan application
  - \* Referral channel
  - \* Satisfaction
  - \* Country
  - \* Region
- 

### 9.1 Customer Type

The dataset contains both individual and corporate customers.

Individual customers represent the majority of the customer base, while corporate customers form a smaller segment.

#### Business Interpretation

The business can therefore consider separate approaches for individual and corporate customers.

Individual customers may require more personalized residential or investment communication, while corporate customers may require business-oriented property solutions.

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### 9.2 Acquisition Purpose

Acquisition purpose is one of the most important variables in the project.

Customers can be broadly understood as purchasing for:

- \* Home/personal use
- \* Investment

This distinction is important because an investment customer may evaluate a property differently from a customer purchasing a home for personal use.

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### 9.3 Loan Application

Loan application information provides an indication of financing requirements.

Customers who apply for loans may benefit from information about financing options and related services.

From a Business Analyst perspective, this variable can help identify opportunities for targeted financial communication.

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#### 9.4 Referral Channel

The customer dataset includes different referral channels, such as:

- \* Website
- \* Agency
- \* Client referral

The website is an important acquisition source.

#### Business Interpretation

This highlights the importance of maintaining a strong digital presence and improving the online customer journey.

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#### 9.5 Satisfaction Score

Customer satisfaction provides an additional perspective on customer experience.

Monitoring satisfaction across different segments can help businesses identify customer groups that may need additional support or improved service.

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### CHAPTER 10: MACHINE LEARNING METHODOLOGY

#### 10.1 Machine Learning Approach

The project uses unsupervised machine learning.

In unsupervised learning, the model does not receive predefined customer categories.

Instead, it identifies patterns and groups customers based on similarities in their data.

This approach is appropriate for the project because the objective is to discover customer segments rather than predict a known target variable.

## 10.2 Methodology

The overall methodology consists of:

### Step 1 – Data Cleaning

Prepare the raw data for analysis.

### Step 2 – Feature Encoding

Convert categorical variables into numerical representations.

### Step 3 – Feature Scaling

Normalize numerical features so that the clustering process is not dominated by a single variable.

### Step 4 – Model Selection

Apply K-Means clustering and consider cluster evaluation methods.

### Step 5 – Cluster Selection

Use the Elbow Method and Silhouette Score to determine an appropriate number of clusters.

### Step 6 – Cluster Interpretation

Analyse each cluster from a business perspective.

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## CHAPTER 11: K-MEANS CLUSTERING

### 11.1 Overview

K-Means is an unsupervised machine learning algorithm used to divide data into a predefined number of groups.

For this project, the final analytical application uses three clusters.

The algorithm works by assigning customers to the nearest cluster centre and repeatedly updating the cluster centres until the grouping becomes stable.

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### 11.2 Why K-Means Was Selected

K-Means was suitable for this project because:

- \* It is relatively simple to implement.

- \* It works well for large datasets.
  - \* It is computationally efficient.
  - \* The resulting clusters are relatively easy to interpret.
  - \* It is widely used for customer segmentation.
- 

## CHAPTER 12: MODEL EVALUATION

### 12.1 Elbow Method

The Elbow Method helps identify an appropriate number of clusters.

The method compares the within-cluster variation for different values of K.

As the number of clusters increases, within-cluster variation normally decreases. The objective is to identify a point where additional clusters provide relatively smaller improvements.

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### 12.2 Silhouette Score

The Silhouette Score evaluates how well customers fit within their assigned clusters.

The Streamlit application produced:

Silhouette Score: 0.4382

The application describes the result as:

Moderate clustering quality

This means the clusters have useful differences, but they are not completely separated.

Therefore, the results should be interpreted as business-oriented customer profiles, rather than rigid customer categories.

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## CHAPTER 13: BUYER SEGMENTATION RESULTS

The project identifies three major customer clusters.

### 13.1 Cluster 0 – Balanced Buyers

Cluster 0 can be interpreted as a relatively balanced customer segment.

These customers do not show an extremely strong concentration toward a single behavioural characteristic.

## Business Characteristics

- \* Balanced customer behaviour
- \* Mixed purchasing motivations
- \* Moderate investment orientation
- \* Mixed financing requirements

## Recommended Approach

The business can use a combination of:

- \* Residential property recommendations
  - \* Investment opportunities
  - \* Financing information
  - \* General property campaigns
- 

### 13.2 Cluster 1 – Senior Buyers

Cluster 1 represents comparatively older buyers.

The age characteristic makes this segment different from the younger customer group.

## Business Characteristics

- \* Older customer profile
- \* Relatively balanced investment behaviour
- \* Different potential financing needs
- \* Greater importance of trust and service

## Recommended Approach

Marketing communication for this group can focus on:

- \* Reliability
  - \* Customer service
  - \* Suitable property options
  - \* Financial clarity
  - \* Long-term value
- 

### 13.3 Cluster 2 – Young Investment-Oriented Buyers

Cluster 2 represents a comparatively younger buyer segment with stronger investment-oriented characteristics.

This makes it particularly relevant for investment-focused marketing.

#### Business Characteristics

- \* Younger customer profile
- \* Relatively stronger investment orientation
- \* Higher relevance for digital engagement
- \* Potential interest in financing and investment opportunities

#### Recommended Approach

The business can target this group with:

- \* Investment-oriented properties
- \* Digital campaigns
- \* Market information
- \* Financing options
- \* Investment education
- \* Portfolio-building opportunities

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## CHAPTER 14: INVESTMENT PROFILING

Investment profiling is an important component of the project.

The purpose is not simply to identify who buys property, but to understand why customers may be acquiring property and what type of support they may require.

The main factors considered include:

#### Age

Age helps identify broad differences between customer groups.

#### Acquisition Purpose

Investment purpose helps distinguish investment-oriented customers from customers purchasing primarily for personal use.

#### Loan Application

Loan behaviour provides information about potential financing requirements.

#### Customer Type

Individual and corporate buyers may have different investment needs.

#### Geography

Country and region help identify geographic patterns.

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## CHAPTER 15: REAL ESTATE PROPERTY ANALYSIS

The project also contains a supporting property/listing dataset.

The property dataset includes information such as:

- \* Listing ID
  - \* Tower number
  - \* Transaction date
  - \* Unit category
  - \* Unit number
  - \* Floor area
  - \* Sale price
  - \* Listing status
  - \* Client reference
- 

### 15.1 Unit Category

The property dataset contains:

- \* Apartments
- \* Offices

Apartments form the larger portion of the property records.

Business Interpretation

The available property data therefore has a strong residential/apartment representation.

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### 15.2 Listing Status

The property data contains properties with different listing statuses, including:

- \* Sold
- \* Available

This information can help a business understand inventory status and sales movement.

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### 15.3 Floor Area

Floor area is an important property characteristic.

It can be used for:

- \* Property comparison
- \* Customer-property matching
- \* Residential analysis
- \* Office property analysis

In a future version of the project, floor area and price information could be combined with customer segments to develop a property recommendation system.

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## CHAPTER 16: DASHBOARD AND VISUALIZATION

### 16.1 Streamlit Application

The project includes a Streamlit application titled:

Real Estate Buyer Segmentation & Investment Profiling

The application uses K-Means clustering for customer segmentation. It provides:

- \* Dataset overview
- \* Customer distribution
- \* Cluster summary
- \* Model evaluation
- \* Elbow Method
- \* Silhouette Analysis
- \* Segmented customer data

The application confirms successful processing of the customer dataset and provides a downloadable segmented customer output.

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### 16.2 Tableau Dashboard

The project also includes a Tableau dashboard structure for presenting customer segmentation information.

Important dashboard views include:

- \* Age Distribution
- \* Client Type Distribution
- \* Cluster Distribution

## \* Customer by Country

These visualizations make it easier for business users to identify patterns without examining raw data directly.

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## CHAPTER 17: BUSINESS ANALYSIS AND INSIGHTS

The project produces several useful business insights.

### 17.1 Different Customers Need Different Strategies

The clustering results demonstrate that the customer base is not completely homogeneous.

Therefore, marketing strategies should be adjusted according to customer characteristics.

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### 17.2 Investment Buyers Are an Important Segment

Investment-oriented customers represent an important business opportunity.

These customers may be more interested in:

- \* Property investment
  - \* Market information
  - \* Growth opportunities
  - \* Financing
  - \* Portfolio development
- 

### 17.3 Young Investment-Oriented Customers Are a Key Target

The younger investment-oriented segment is particularly useful for targeted campaigns.

Digital communication can be especially relevant for this group.

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### 17.4 Financing Behaviour Can Support Targeting

Loan application behaviour provides useful information for identifying customers who may need financing support.

This can help businesses connect customers with appropriate financial products and information.

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## 17.5 Digital Acquisition Is Important

The website represents a significant customer acquisition channel.

This suggests that improving the digital customer journey can support lead generation and customer engagement.

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## 17.6 Customer Satisfaction Should Be Monitored

Customer satisfaction can be used alongside segmentation to identify whether particular groups are experiencing better or weaker customer experiences.

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## CHAPTER 18: RECOMMENDATIONS

Based on the analysis, the following recommendations are proposed.

### 18.1 Use Segment-Based Marketing

The company should avoid using exactly the same communication strategy for every customer.

Marketing campaigns should be customized according to customer segments.

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### 18.2 Target Investment-Oriented Customers

Investment-oriented customers should receive specialized information related to:

- \* Investment opportunities
  - \* Property market trends
  - \* Potential returns
  - \* Financing
  - \* Portfolio development
- 

### 18.3 Strengthen Digital Marketing

The importance of the website as an acquisition channel suggests that the organization should continue improving:

- \* Website usability
- \* Online property search
- \* Digital advertising

- \* Lead generation
  - \* Online customer engagement
- 

#### 18.4 Use Financing Information

Customers showing financing requirements can be provided with appropriate information about available financing options.

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#### 18.5 Personalize Property Recommendations

Customer clusters can be connected with property characteristics to create more personalized recommendations.

For example:

Young Investment-Oriented Buyer → Investment-focused property options

Balanced Buyer → Residential + investment options

Senior Buyer → Suitable long-term/value-focused properties

These recommendations should be further validated using actual transaction and preference data.

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#### 18.6 Monitor Customer Satisfaction

Satisfaction should be tracked across customer segments.

If a particular segment shows lower satisfaction, the organization can investigate the reasons and improve the customer experience.

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### CHAPTER 19: LIMITATIONS

Every analytical project has limitations, and this project is no exception.

#### 19.1 Moderate Clustering Quality

The Silhouette Score of 0.4382 indicates moderate clustering quality.

This means the clusters are useful for analysis but are not perfectly separated.

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## 19.2 Dataset Limitation

The project uses the available project dataset. Therefore, the findings should not automatically be considered representative of the entire real estate market.

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## 19.3 No Causal Interpretation

The clustering model identifies similarities between customers.

It does not prove that one customer characteristic causes another behaviour.

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## 19.4 Separate Data Sources

The customer segmentation data and property data should only be directly combined when a valid relationship between their identifiers has been established.

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## 19.5 Feature Selection

The quality of a clustering model can be affected by which variables are included, how categorical variables are encoded and how features are scaled.

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# CHAPTER 20: FUTURE SCOPE

The project can be improved further in several ways.

## 20.1 Compare More Machine Learning Algorithms

Future analysis can compare:

- \* K-Means
- \* Hierarchical Clustering
- \* DBSCAN
- \* Gaussian Mixture Models

This can help identify whether another method provides better segmentation.

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## 20.2 Use PCA

Principal Component Analysis can be used to reduce the dimensionality of the data and visualize customer clusters more clearly.

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### 20.3 Customer-Property Recommendation

Customer clusters can be connected with property characteristics to create a recommendation system.

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### 20.4 Real-Time Business Dashboard

The Streamlit application can be expanded with interactive filters for:

- \* Country
  - \* Region
  - \* Client type
  - \* Acquisition purpose
  - \* Cluster
  - \* Property category
- 

### 20.5 Predictive Analytics

Future versions could include predictive models for:

- \* Investment probability
- \* Loan requirement
- \* Customer conversion
- \* Property preference
- \* Customer churn

Such models would require appropriate target variables and validated historical data.

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### 20.6 Automated Business Reporting

The system could automatically generate weekly or monthly reports showing:

- \* Customer segment movement
- \* Investment trends
- \* Lead sources
- \* Customer satisfaction
- \* Property demand
- \* Sales performance

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## CHAPTER 21: CONCLUSION

The project “Machine Learning Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence” demonstrates how business analysis and machine learning can work together to solve a practical business problem.

The project starts with customer data and applies data preparation, exploratory analysis, feature encoding, feature scaling and K-Means clustering.

The final analytical application processes 2,000 customer records with 20 features and uses 3 selected clusters.

The model produces a Silhouette Score of 0.4382, which indicates moderate clustering quality.

The resulting segments provide useful business profiles, including balanced buyers, senior buyers and younger investment-oriented buyers.

From a Business Analyst perspective, the main value of the project is not only the machine learning model itself. The important outcome is the ability to translate analytical results into business actions.

The segmentation can support:

- \* More focused marketing
- \* Better customer targeting
- \* Investment profiling
- \* Financing-related communication
- \* Customer relationship management
- \* Property recommendations
- \* Data-driven decision-making

The project also demonstrates how dashboards and visualizations can make analytical information easier for business users to understand.

Overall, the project provides a practical example of how Business Analytics + Machine Learning + Visualization can be combined to generate useful insights for real estate market intelligence.

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## APPENDIX A: PROJECT TECHNOLOGY STACK

| Technology                     | Purpose                               |
|--------------------------------|---------------------------------------|
| Microsoft Excel                | Data storage and preliminary analysis |
| Python                         | Data analysis and machine learning    |
| Google Colab                   | Model development                     |
| Pandas                         | Data manipulation                     |
| Scikit-learn                   | Machine learning                      |
| K-Means                        | Customer segmentation                 |
| Streamlit                      | Interactive application               |
| Tableau                        | Dashboard and visualization           |
| Matplotlib/Visualization Tools | Charts and graphical analysis         |

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## APPENDIX B: PROJECT WORKFLOW

Business Problem

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Data Collection

↓

Data Cleaning

↓

Exploratory Data Analysis

↓

Feature Encoding

↓

Feature Scaling

↓

K-Means Clustering



Elbow Method & Silhouette Evaluation



Customer Segmentation



Investment Profiling



Dashboard Visualization



Business Insights



Recommendations



## APPENDIX C: FINAL MODEL SUMMARY

### Metric Result

Project Buyer Segmentation & Investment Profiling

Internship Organization Unified Mentor

Internship Domain Business Analyst

Dataset Records 2,000

Features 20

Selected Clusters 3

Machine Learning Method K-Means Clustering

Evaluation Metric Silhouette Score

Silhouette Score 0.4382

Clustering Quality Moderate

Application Streamlit

Visualization Tableau



## FINAL PROJECT SUMMARY

Organization: Unified Mentor

Internship: Business Analyst

Project: Machine Learning Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence  
Domain: Financial Analytics & Real Estate Market Intelligence  
Dataset: 2,000 customer records  
Features: 20  
Model: K-Means Clustering  
Clusters: 3  
Silhouette Score: 0.4382  
Result: Moderate clustering quality  
Application: Streamlit  
Dashboard: Tableau

The project demonstrates the practical use of business analysis and machine learning to understand real estate customers and support data-driven business decisions.